

Rishu Sharma

cyber hygiene

 Cyber hygiene

Document Details

Submission ID**trn:oid::3618:117502036****Submission Date****Oct 19, 2025, 11:40 PM GMT+5:30****Download Date****Oct 19, 2025, 11:42 PM GMT+5:30****File Name****cyber hygiene.docx****File Size****238.8 KB****15 Pages****3,578 Words****23,109 Characters**

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **16 Not Cited or Quoted 4%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 1%  Internet sources
- 2%  Publications
- 2%  Submitted works (Student Papers)

Match Groups

- 16 Not Cited or Quoted** 4%
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations** 0%
Matches that are still very similar to source material
- 0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 1% Internet sources
- 2% Publications
- 2% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

- 1** **Publication**
Celestine Ugwu, Casimir Ani, Modesta Ezema, Caroline Asogwa et al. "Towards De... 1%
- 2** **Student papers**
Inu on 2024-01-26 <1%
- 3** **Internet**
dumpsboss.com <1%
- 4** **Student papers**
American Public University System on 2024-11-24 <1%
- 5** **Publication**
Iim Halimatusa'diyah, Ella Prihatini. "Young voters and political participation in I... <1%
- 6** **Internet**
chinadailyshow.com <1%
- 7** **Internet**
repository.up.ac.za <1%
- 8** **Student papers**
Nottingham Trent University on 2024-04-18 <1%
- 9** **Publication**
Tuğçe Karayel, Bahadır Aktaş, Adem Akbıyık. "Human factors in remote work: exa... <1%

CYBER HYGIENE

Table of Content

1. Introduction.....	2-5
2. Methodology.....	5-7
2.1 Instrument and Method of Data Collection	5-6
2.2 Sampling Technique.....	6-6
2.3 Research Model.....	6-7
2.4 Research Hypotheses.....	7-7
2.5 Data Analysis Procedure.....	7-7
3. Results.....	7-12
3.1 Devices Used for Internet Access.....	7-8
3.2 Purpose of Internet Usage.....	8-9
3.3 Demographic Distribution.....	9-9
3.4 Age and Cyber Hygiene.....	9-10
3.5 Educational Level and Cyber Hygiene.....	10-11
3.6 Overall Cyber Hygiene Culture.....	11-12
4. Cyber Hygiene best practises.....	12-13
5. Conclusion.....	13-16
5.1 General Findings.....	13-14
5.2 Influence of Age on Cyber Hygiene.....	14-14
5.3 Effect of Educational Level on Cyber Hygiene.....	14-14
5.4 Behavioral Patterns and Use of Security Tools.....	14-14
5.5 Gender Differences in Cyber Hygiene.....	15-15
5.6 Impact of previous Cyber attacks.....	15-15
5.7 Role of Cybersecurity Training.....	15-15
5.8Implications.....	15-15
5.9Sumaary and Key insights.....	16-16
6. References.....	16-16

CYBER HYGIENE

1. Introduction

Statement:-

Evolution of Cyber Threats and Impact on Sectors

When the IT industry first began to expand, early attacks were simple spam messages, which eventually grew into more dangerous threats like **computer viruses and malware**, including the infamous **WannaCry** incident. The increasing number of cybersecurity breaches has become a major worry, particularly for the **healthcare sector**. While cybersecurity is a global issue, healthcare organizations have been slower to adopt strong data protection measures compared with other sectors.

With the quick shift toward digital patient records, hospitals have suffered significant **financial and reputational damage** from data breaches. Between 2015 and 2016, phishing and social engineering attacks on healthcare organizations saw the steepest annual increase. Cyber-attacks now impact not only corporations but also individuals, who face growing risks of personal loss. In 2015 alone, Americans filed around **288,000 cybercrime complaints**, with over 40% of those cases resulting in monetary losses.

According to 2015 law enforcement reports, hundreds of thousands of cybercrime complaints were filed in the United States, leading to massive financial losses totaling **over a billion dollars**. The average loss per incident was several thousand dollars, highlighting the real economic consequences of cybercrime.

User Vulnerability and Awareness Gaps

End users are still seen as the most vulnerable point in cybersecurity, especially in **personal computing environments**, where they are the primary targets. This is because home systems typically lack dedicated IT professionals responsible for timely maintenance and updates of software and hardware. Consequently, even highly secure systems can be compromised if users practice unsafe digital habits, such as sharing sensitive details online or neglecting basic protective measures.

Although media coverage has made people aware of potential risks, most still struggle to apply best practices, including **password management and secure browsing**. Despite the wide availability of security tools, many users remain unsure how to select, configure, or maintain them. Surveys have uncovered significant gaps in cybersecurity knowledge: a large number of participants couldn't tell the difference between an **antivirus program and a firewall**. About two-thirds of respondents lacked updated antivirus protection, and nearly three-fourths had misconfigured or inactive firewalls. In another extensive survey, 70% of users relied on antivirus software at home, 70% used a firewall, 40% employed anti-phishing tools, 75% had anti-spyware software, and fewer than 20% utilized intrusion detection systems.

Longitudinal studies show that specific groups—especially **software developers and gamers**—face a higher frequency of malware attacks than casual users. Keeping security software current is thus essential. Risk perception studies suggest that individuals who feel more in control of their devices and who understand the severity of potential consequences are more likely to take precautionary steps, like updating security tools. Interestingly, **men were found to be more likely than women to perform regular updates**.

Shifting Focus to Human Factors and Research Findings

Recent reviews emphasize a major shift in cybersecurity threats from targeting IT infrastructure to **exploiting human vulnerabilities**. For example, healthcare professionals' use of social media has become an effective method for launching targeted phishing attacks. Research by Jalali and Kaiser indicated that **employee workload** influences risky behaviors, such as clicking on malicious links. To address these weaknesses, experts have recommended developing training modules that promote privacy awareness and proper use of social media settings. Furthermore, they stressed the need for structured organizational programs to assess cybersecurity risks and human factors, particularly in the healthcare sector, where digital transformation has outpaced user training.

A study investigating user knowledge and behaviors related to cyber hygiene recruited over 300 participants, with data from about 270 analyzed after excluding invalid responses. Participants completed an online survey measuring their knowledge of cybersecurity concepts, awareness of threats, and actual behaviors across topics like **authentication, phishing, social networking, mobile security, and safe browsing**.

The study found a clear disconnect between knowledge and behavior. While more than 90% of participants created passwords of at least eight characters, only 70% included special symbols, and nearly **half reused passwords** across multiple accounts. Alarming, only **5% refrained from sharing passwords**. About half of the respondents admitted to risky behaviors such as opening links or attachments from unknown senders. Although 80% reported using spam filters, few participants actively changed or strengthened their passwords regularly.

Overall, the findings confirm that **cyber hygiene awareness doesn't always translate into safe online practices**. Demographic factors such as age, gender, and prior attack experience influence the likelihood of engaging in protective behaviors. The results underscore the need for **tailored cybersecurity education** and user-friendly awareness programs that effectively bridge the gap between knowing what to do and actually doing it.

DEVICES USED FOR INTERNET

Variable	Frequency	Percentage
Laptop	205	35.65
Tablet	42	7.30
Desktop	27	4.7
Other devices	12	2.80
Mobile phone	289	86.70

The Human Element: The Weakest Link in Cybersecurity

The **human element** remains one of the most significant vulnerabilities in cybersecurity. Despite heavy investment in technical defenses like **firewalls, encryption, and network security**, end-users frequently act as the weakest link in the security chain.

The Problem: Poor Cyber Hygiene

Despite increasing awareness of data protection and online risks, many users continue to exhibit **poor cyber hygiene** behaviors, such as:

- Using simple passwords.
- Sharing credentials.
- Ignoring crucial software updates.

These habits create easy-to-exploit gaps for attackers who prefer to infiltrate systems through **human error** rather than complex technical flaws.

The New Paradigm: Exploiting Behavior

Cybercriminals are increasingly exploiting this behavioral weakness by shifting their focus from attacking **infrastructure** to **exploiting human vulnerability**. This represents a major paradigm change in cybersecurity.

- Instead of relying on developing complex malware, attackers now heavily use **social engineering** and **psychological manipulation** to compromise user accounts.
- Effective defense, therefore, requires addressing **not only technical** resilience **but also user awareness, education, and behavioral change**.

User Vulnerability and Awareness Gaps

Studies consistently show a significant gap between users' awareness of cybersecurity risks and their application of appropriate protection methods.

- **Targeting Individuals:** Because home and personal computing devices are rarely monitored by dedicated IT professionals, they are often left unpatched and vulnerable. As a result, nearly **95% of cyber-attacks** target individuals rather than institutions. These attacks frequently exploit outdated software, weak passwords, or poor browsing practices.
- **Misconfiguration and Lack of Updates:** Despite the availability of security software, many users lack the understanding needed to configure or maintain them.
 - Surveys indicate that nearly **two-thirds of users** either lack updated antivirus software or have none at all, or operate systems with **misconfigured firewalls**.
 - Users often **confuse the functions** of security tools, mistakenly believing that antivirus protection alone is sufficient.

- **Data on Protective Measures:** Another large-scale behavioral study found that:
 - While a majority of untrained users had antivirus and firewall software, **less than 20%** had implemented **intrusion detection systems**.
 - Alarmingly, nearly **three-fourths of users** did not regularly update their protective software, leaving their systems susceptible to known vulnerabilities.

Age Group	Frequency	Percentage
15-24	80	25.3
25-34	120	38
35-44	65	20.6
45-54	40	12.6
55+	11	3.5

Table:- shows the usage by age

Human-Centric Vulnerability in Healthcare

Nifakos and colleagues (2023) observed a growing shift in cybersecurity research, now emphasizing **human behavior** as a critical vulnerability within healthcare organizations, rather than purely technical flaws.

- Jalali and Kaiser demonstrated that **workload pressure** and **cognitive fatigue** can increase the likelihood of employees clicking malicious links or neglecting security policies.
- These findings support the need for **human-centered cybersecurity strategies**, which focus not only on systems but also on users' **awareness, workload, and behavior**.
- Organizations are now being urged to conduct **cyber risk assessments** and **privacy impact evaluations** to identify behavioral vulnerabilities alongside technical ones.

Training and Standardization Gaps

To mitigate these risks, experts have proposed implementing **training modules and awareness campaigns tailored to healthcare professionals**. These programs encourage the proper use of privacy settings and responsible handling of patient data. However, there is currently **little consensus** on standardized cybersecurity training curricula or systematic evaluation methods for improving cyber hygiene (CH) among healthcare workers.

Empirical Study: The Knowledge-Action Gap

A recent empirical study recruited over 300 participants via Amazon Mechanical Turk to assess their cybersecurity knowledge and behavior. After excluding invalid responses, the dataset included approximately 270 participants, with a balanced gender distribution and a wide age range (18 to 55

years). Participants completed an online questionnaire evaluating their knowledge of cyber concepts, awareness of threats, and self-reported security behaviors across domains like **authentication, phishing awareness, social networking, mobile device safety, and general digital hygiene**.

Key Findings on Behavior

The results revealed a **clear gap between awareness and behavior**:

- **Password Habits:** While 90% claimed to create passwords with **at least eight characters and use both uppercase and lowercase letters**, only **70%** incorporated special symbols, and nearly **half reused the same password** for multiple accounts. Alarming, only **5%** stated they never share their passwords.
- **Phishing Behavior:** Findings here were even more concerning: over **90% of participants admitted to opening links or attachments from unknown senders**, and almost all had, at some point, responded to suspicious emails. Although most reported using spam filters, few participants routinely verified the legitimacy of senders or practiced secure communication protocols.

Conclusion: Why Awareness Fails

The data confirms that cyber hygiene knowledge does not automatically lead to secure behavior. Many users are aware of risks but fail to apply this understanding consistently. Factors such as **convenience, cognitive overload, and underestimation of personal risk** contribute to unsafe digital practices. The gap between knowledge and action emphasizes the need for **interactive and practical cybersecurity education**, rather than passive awareness campaigns.

Mitigation Strategies

Organizations should adopt a **multi-layered defense model** that integrates technical controls with behavioral interventions:

- **Regular awareness and phishing simulation training.**
- **Automatic software update enforcement.**
- **Clear password management policies.**
- **Behavior-based monitoring systems.**

2. Methodology

2.1 Instrument and Method of Data Collection

The study utilized a questionnaire as its primary instrument, distributed to respondents through multiple channels, including the institution's official academic groups. This approach aimed to ensure a diverse range of participants across various departments and levels of education.

The questionnaire demonstrated **strong reliability**, confirmed by a **Cronbach's alpha score of $\alpha = 0.86$** . This score indicates that the items had high internal consistency and were suitable for consistently measuring the intended constructs. Prior to full deployment, the instrument underwent content validation by subject matter experts and a pilot test to verify its clarity and reliability.

2.2 Sampling Technique

The research employed **convenience sampling**, chosen for its practicality and affordability within an academic setting, allowing for the timely collection of online responses from students and staff.

The sample achieved a **wide demographic spectrum**, including participants from diverse age groups (15–54+ years) and educational backgrounds (from diploma holders to Ph.D. candidates). This diversity was crucial for examining variations in cyber hygiene knowledge and behavior across different age and education levels.

2.3 Research Model

The study model considered two **independent demographic variables**:

1. **Age**
2. **Educational level**

The single **dependent variable** was **Cyber Hygiene**, defined as the set of regular practices and behavioral patterns that maintain the security, integrity, and confidentiality of information systems.

Four specific **dimensions of Cyber Hygiene** were measured:

1. **Social Network Cyber Hygiene (SNCH)**: Focuses on safe online interactions, privacy management, and responsible sharing practices on social platforms.
 2. **Storage and Virus Attack Cyber Hygiene (SVCH)**: Involves the use of antivirus software, data backup habits, and malware prevention.
 3. **Authentication Cyber Hygiene (ACH)**: Measures password management, the use of two-factor authentication, and login security practices.
 4. **Social Engineering Cyber Hygiene (SECH)**: Assesses awareness and resistance to phishing, scams, and manipulative online tactics (a dimension newly introduced in this study).
-

2.4 Research Hypotheses

The investigation was guided by the following **null hypotheses (\$H_0\$)**, which were formulated to be tested statistically:

- **\$H_0\$ (Age)**: Age of the internet user has no significant effect on cyber hygiene behavior.
- **\$H_0\$ (Education)**: Educational level of the internet user has no significant effect on cyber hygiene behavior.

These hypotheses were tested using **chi-square** and **ANOVA** analyses.

2.5 Data Analysis Procedure

Data obtained via Google Forms was exported to **SPSS** for statistical analysis, involving data cleaning, coding, and categorization. The questionnaire was structured into three analytical sections:

Section	Content Focus	Scale Used
Section A	Demographic information, device usage, and online activity patterns.	N/A (Categorical/Nominal)
Section B	Knowledge of cybersecurity threats and concepts (12 questions).	Four-point Likert Scale (Strongly Agree to Strongly Disagree)
Section C	Behavioral engagement in cyber hygiene practices.	Four-point Likert Scale (Every time to Never)

3. Results

3.1 Devices Used for Internet Access

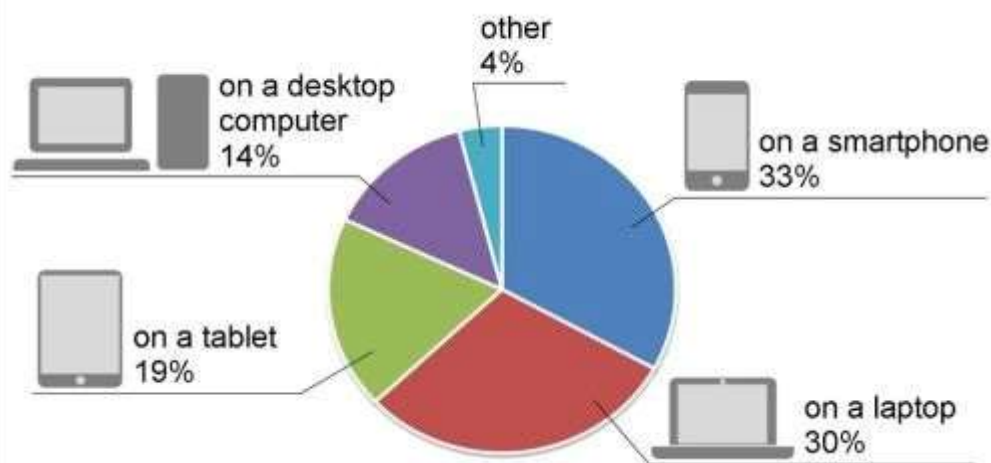
An analysis of device usage among respondents revealed that the **mobile phone** 📱 was the most commonly used device for internet access, accounting for **289 (50.26%)** of the responses.

This was closely followed by **laptops** 💻, which were used by **205 (35.65%)** of the participants.

The high popularity of mobile phones reflects factors such as their **affordability, accessibility, and portability** compared to less mobile options like laptops. This usage pattern aligns with broader national trends, though it slightly **contrasts** with some other studies where laptops held the dominant position for internet access.

The pie chart below illustrates how internet users aged 16+ prefer to access the internet at home and in other places.

Summarise the information in the pie chart by selecting and reporting the main features.



3.2 Purpose of Internet Usage-

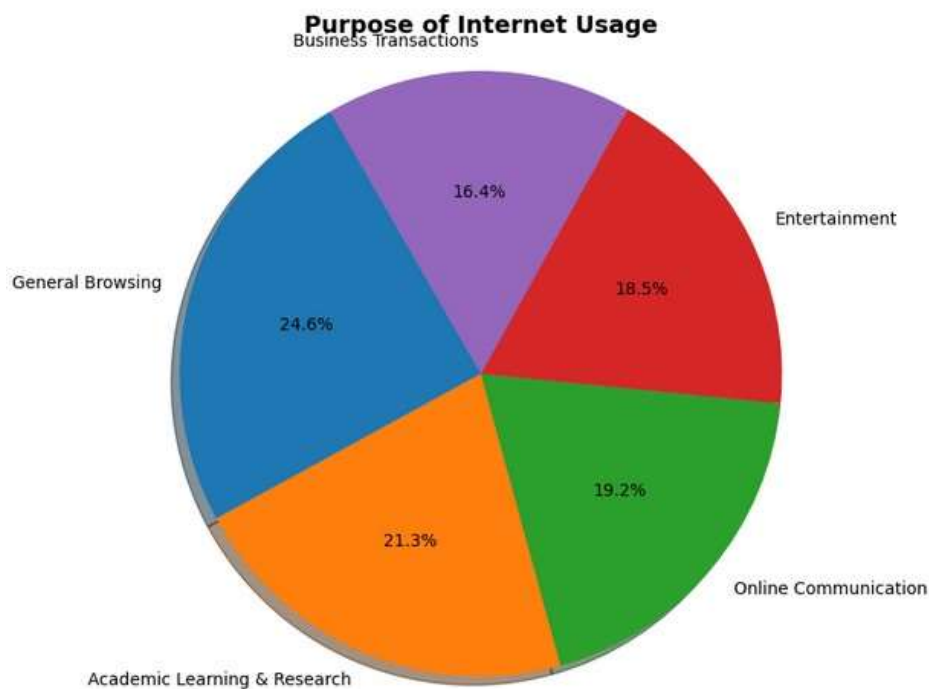
The most frequent use of the internet among the participants was **general browsing**, accounting for **17.97%**.

This was closely followed by **academic learning and research**, at **15.59%**.

Other significant uses included:

- Online communication.
- Entertainment.
- Business-related transactions.

These results suggest that the majority of respondents engage online for **educational purposes**, underscoring the internet's increasing importance in digital learning environments.



3.4 Age and Cyber Hygiene

Statistical results indicated **no meaningful link between age and cyber hygiene habits** in this study. This outcome aligns with earlier research while challenging the popular assumption that younger individuals inherently maintain stronger cybersecurity awareness.

Educational Demographics

The study sample was dominated by younger, academically active individuals, who are likely more exposed to digital technologies. The educational distribution was as follows:

- **Undergraduates:** 45.9%
- **Master's degree holders:** 22.5%
- **Graduates:** 15.5%
- **Doctoral candidates:** 15.5%

Statistical Findings

A Chi-square test was performed to examine the relationship between age and the participants' cyber hygiene culture.

- The result yielded a **p-value of \$0.649\$, which is greater than the** standard significance level.
- Therefore, the **null hypothesis** (\$H_0\$: Age has no significant effect on cyber hygiene behavior) was **accepted**.

Internet Activities by Age Group

	Age group						
Activity%	Teens	20s	30s	40s	50s	60s	70+
Get News	76	73	76	75	71	74	70
Online games	81	54	37	29	25	25	32
Downloads	52	46	27	15	13	8	6
Product research	0	79	80	83	79	74	70
Buying a product	43	68	69	68	67	65	41
Searching for people	5	31	23	23	24	29	27

3.5 Educational Level and Cyber Hygiene

The analysis of the **educational level** variable against cyber hygiene culture, using **the Chi-square test**, showed **no significant** relationship.

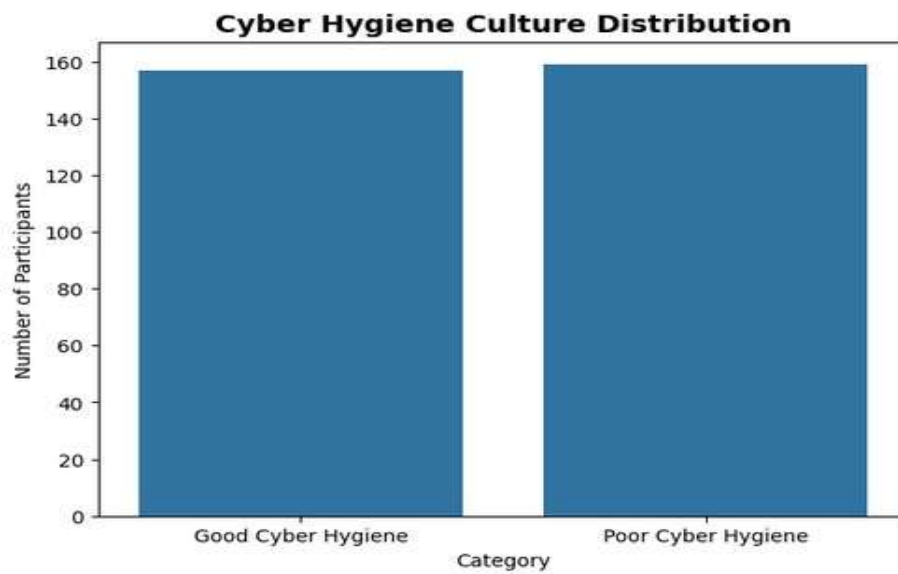
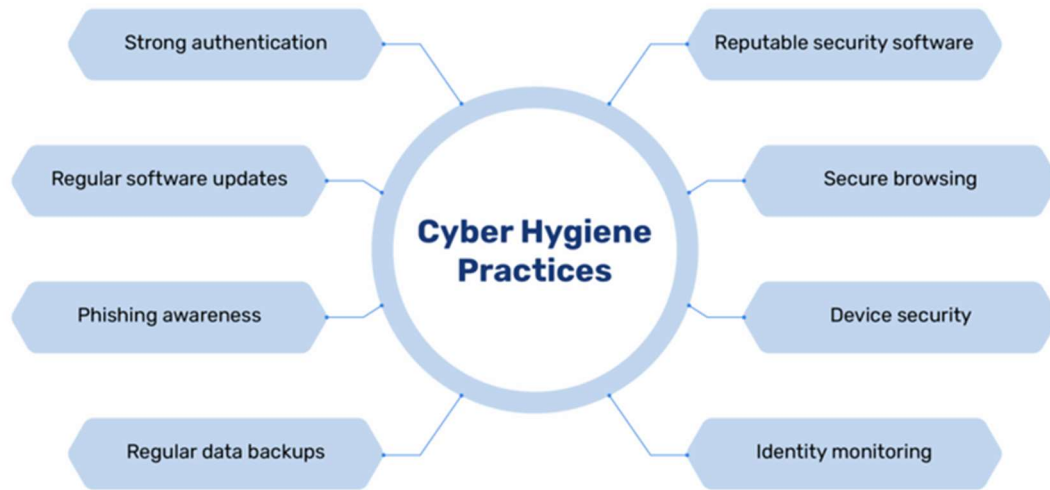
The statistical test yielded a **p-value of \$0.337\$**, which is **greater than the** standard \$0.05\$ significance threshold. Consequently, **the null hypothesis—that** educational level has **no significant** effect on cyber hygiene behavior—was **accepted**.

Total cyber-hygiene knowledge score	Statistic	Std. Error
Mean	9.39	0.123
95% confidence upper range	9.15	
Interval of mean lower range	9.64	
5% Trimmed Mean	9.49	
Median	10.00	
Variance	2.20	
Std. Deviation	1.48	
Minimum	4.00	
Maximum	12.00	
Range	8.00	
Interquartile Range	2.00	
Skewness	-0.87	0.201
Kurtosis	0.81	0.400

3.6 Overall Cyber Hygiene Culture

To determine the general state of cyber hygiene among participants, their scores were analyzed relative to the computed mean.

- The analysis revealed a near-equal distribution:
 - 50.7%** of respondents **were classified as having a High Cyber Hygiene Culture.**
 - 49.3%** of respondents **were classified as having a Low Cyber Hygiene Culture** (i.e., displayed poor cyber hygiene culture).



4. Cyber Hygiene best practises:-

1. Credential and System Protection

- **Keep Your Passwords Secret:** Your password is your first line of defense. **Never share it** with anyone unless absolutely necessary. Create **strong passwords that combine uppercase and lowercase letters, numbers, and symbols.** **Change them regularly and avoid reusing** the same password for multiple accounts to limit damage from a data breach.
- **Install and Update Antivirus Software:** Use reliable antivirus software to detect, quarantine, and remove malware before it causes harm. Always **keep your antivirus updated** so it can recognize the latest threats. For extra defense, pair it with a **firewall** to block unauthorized network access.
- **Encrypt Storage Devices:** Always **encrypt your computer's hard drive and any external storage devices.** This **ensures that even if they're lost or stolen, your data remains inaccessible to unauthorized users.** Full-disk encryption is especially vital **for** portable devices like laptops.
- **Lock Your Screen When Away:** Make it a habit to **enable a password-protected screensaver** whenever you step away from your system. This simple step prevents unauthorized people from accessing your data, particularly in shared or public areas.

2. Data Integrity and Incident Response

- **Maintain Regular Backups:** Always **back up your important files** to secure storage. Use a combination of **cloud servers and offline external drives.** Regular backups mean you can recover data quickly in the event of ransomware, hardware failure, or accidental deletion.
- **Report Any Suspicious Activity Immediately:** If you notice unusual system behavior—such as unauthorized logins, unexpected file changes, or new software—**report it immediately** to your IT administrator or cybersecurity team. Quick reporting helps security teams investigate and prevent further data loss or compromise.

3. Safe Online Conduct

- **Avoid Suspicious Websites:** **Refrain from visiting unverified or suspicious websites;** they often host spyware or malware designed to steal your data. Pay close attention to any warnings issued by your modern browser. If you find a malicious link, report it to the proper authority for review.
- **Use Secure Email Practices:** Treat emails as inherently insecure, much like an open postcard. **Use encryption tools** when sending confidential data. Be extremely **cautious with attachments and links,** as phishing emails are still one of the most common methods attackers use to steal information.

4. Physical Security

- **Secure Physical Documents and Media:** Don't forget physical security! **Avoid leaving printed documents, USB drives, or external hard disks unattended** on desks where others can access them. Store them securely in **locked drawers or cabinets** when not in use.
- **Remove Sensitive Documents from Shared Devices:** When using shared office equipment like printers, fax machines, or photocopiers, **promptly collect your documents** after they finish printing or scanning. Leaving sensitive paperwork behind dramatically increases the risk of information leakage or theft.

5.CONCLUSION

- The study systematically analyzed cyber hygiene among internet users, concluding that **demographic factors like age and educational level do not significantly predict** good security practices. Good cyber hygiene is instead driven by **individual awareness, motivation, and exposure to effective training.**

- **5.1 General Findings**
- **Mobile Dominance:** The **mobile phone** is the most frequent device for internet access (approximately half of participants), indicating a strong shift toward **portable and on-the-go connectivity**. This introduces new security risks if users neglect essentials like device encryption and updates.
- **Activity Focus:** Most respondents primarily engage in **browsing and learning-related tasks**, highlighting the internet's critical role in e-learning and research. This dependence amplifies the importance of secure online behavior, particularly when handling sensitive academic data.
- **Poor Hygiene Culture:** A significant proportion of surveyed individuals—**about half**—displayed **inadequate cyber hygiene**, exhibiting unsafe behaviors such as weak passwords, failure to update software, and carelessness with confidential information. This underscores the pressing need for **targeted awareness programs.**

- **5.2 Influence of Age and Education**
- **Age Insignificance:** The Chi-square test showed **no meaningful association between age and cyber hygiene ()**. This challenges the common assumption that being younger automatically equates to better digital literacy or security behavior. In fact, some **older participants demonstrated more cautious online behavior**, potentially due to higher risk awareness.
- **Education Insignificance:** Similarly, **educational level was statistically insignificant ()** in influencing cyber hygiene practices. Formal academic qualification may not emphasize practical digital safety habits unless they are specifically taught. This calls for an **integrated approach** where cyber hygiene is embedded into academic curricula.

- **5.3 Behavioral Patterns and Security Tools**
- **Antivirus Usage Gap:** While a majority of participants reported installing and regularly updating antivirus software, only a smaller proportion performed **frequent system scans**. This illustrates a common gap between **awareness and consistent action.**
- **Firewall Misuse:** Between and of participants used a firewall, but most **rarely altered the configurations** to optimize protection. This suggests users often lack the **technical knowledge** to use security tools effectively.
- **Social Media Risk:** Despite using anti-spyware software (more so among older users), **both age groups shared excessive personal information** (phone numbers, addresses, photos) on social media and neglected to review their privacy settings, making them vulnerable to identity theft and phishing.

- **5.4 Other Influencing Factors**

- **Gender:** Male participants showed slightly higher cyber hygiene knowledge, but **behavioral differences between genders were not statistically significant**. Men's higher expressed confidence did not translate into markedly different practices compared to females.
- **Prior Attacks:** Surprisingly, **having experienced a prior cyber attack did not significantly influence** participants' current cyber hygiene practices. Victims of past incidents continued to display inconsistent protective habits, indicating a **lack of learning reinforcement** or insufficient post-incident education.
- **Training Shortcomings:** While of respondents reported receiving some form of cybersecurity training, this exposure **did not lead to significant improvement** in either knowledge or behavior. Current programs often emphasize theoretical knowledge but **fail to instill habitual, behavior-based learning**.

- **5.5 Implications and Future Directions**

- The study emphasizes that cyber hygiene must be treated as a **continuous behavioral discipline**, not a one-time initiative.
- **Intervention Strategy:** Institutions must prioritize **habit-based cybersecurity interventions**, using tools like periodic reminders, gamified assessments, and reward mechanisms to reinforce safe practices.
- **Future Research:** Future studies should explore other potential factors influencing cyber hygiene, such as gender identity, socio-economic background, cultural context, and the type of internet usage. **Longitudinal studies** are also needed to track the long-term impact of sustained training on user behavior.